



Medical Microbiology

Lab. 3

Culture Media in Microbiology

Third stage- College of Dentistry

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What is culture media?

- Culture media are nutrient-rich substances used to support the growth of microorganisms outside the body (in vitro).
- These preparations provide the necessary environment for microorganisms to grow and multiply in laboratory conditions.
- The substance itself is called the **culture medium**, and the microbial growth that occurs within it is referred to as a **culture**.

Uses of Culture Medium:

- To identify the causative agent of an infection from clinical samples, enabling appropriate treatment.
- To study the characteristics and behavior of microorganisms.
- To produce biological products such as vaccines, toxoids, antigens, and more.

Composition of culture media:

- 1- Water.
- 2- Energy source.
- 3- Carbon source.
- 4- Nitrogen source.
- 5- Mineral salts.
- 6- Special growth factors .



Types of Culture Media based on physical state:

1-Solid Media:

These media are solid in form and are used for the isolation and cultivation of microorganisms.

- **Agar** is the primary solidifying agent, typically added at a concentration of **1.5%**.
- **Examples:** Blood agar, MacConkey agar.

Types of culture media



2-Semi-Solid Media:

These have a jelly-like consistency and are prepared by adding a small amount of agar (**0.2–0.5%**) to a liquid medium. They are often used for motility testing.

- **Example: Sulfide, Indole, Motility** medium (SIM medium).

3-Liquid Media (Broth):

These media remain in a liquid state and do not contain agar. They are commonly used for growing large volumes of bacteria.

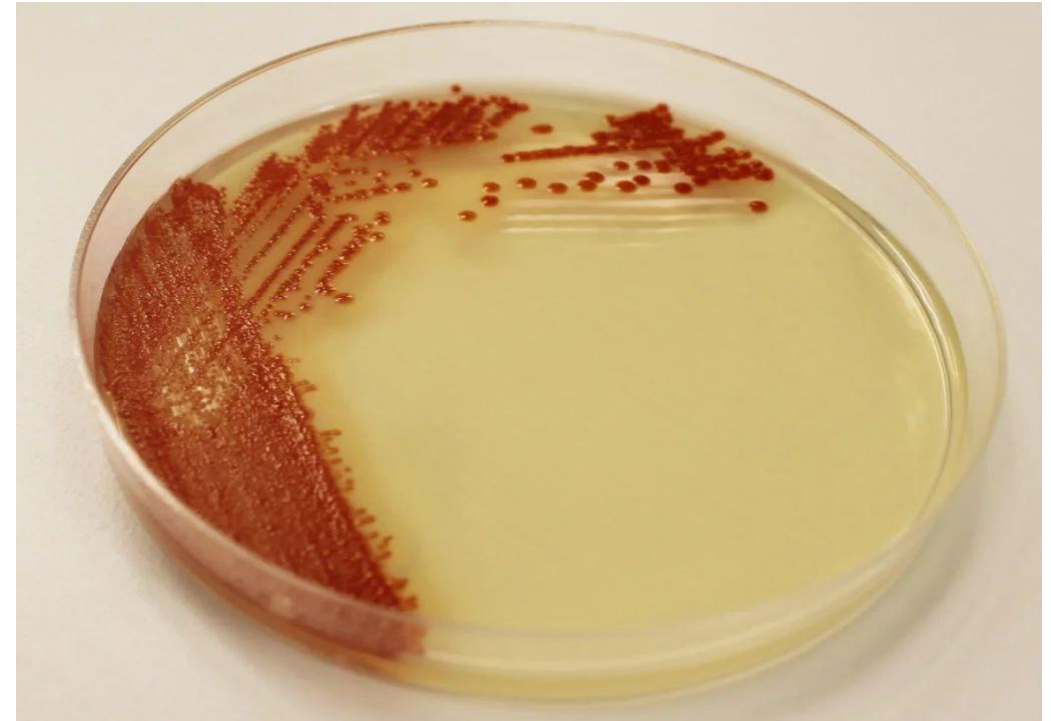
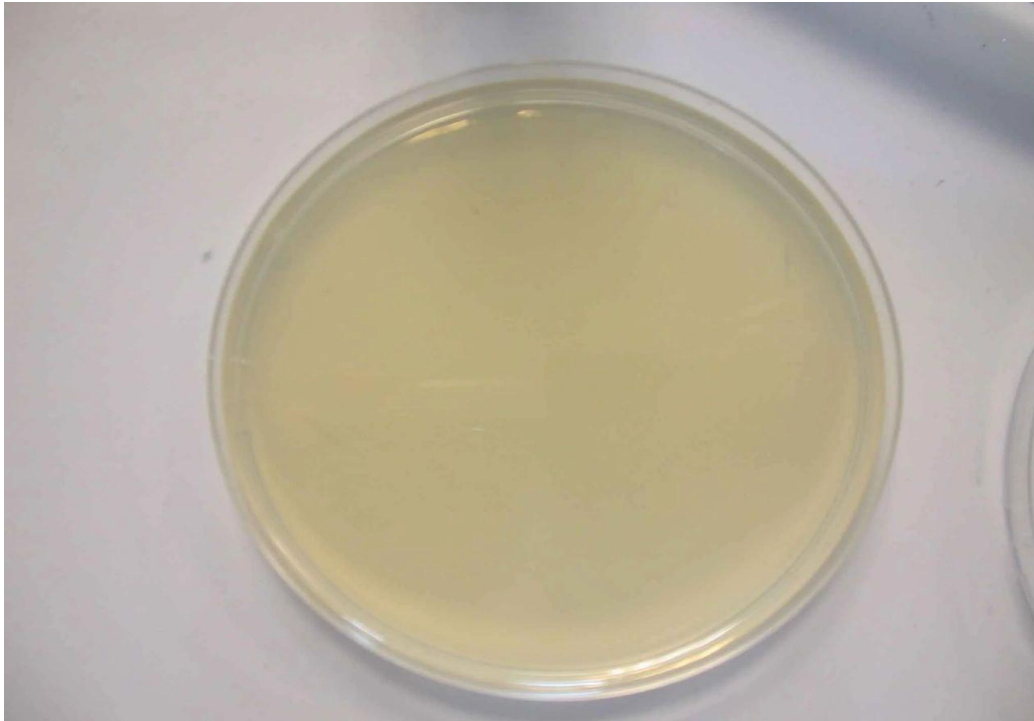
- **Examples:** Nutrient broth, MacConkey broth.



B . Classification based on the ingredients:

A- Simple media :

Examples : Nutrient Agar



B- Complex media :

Examples : Blood agar



C-Special media:

- 1- Enriched media
- 2- Selective media
- 3- Differential media
- 4- Transport media
- 5- Anaerobic media

1- Enriched media:

Substances like blood , serum, egg are added to the simple medium,used to grow bacteria.

Examples, Chocolate agar, Blood agar.



2. Selective Media:

- Selective media contain specific components that **inhibit** the growth of certain types or species of microorganisms while **promoting** the growth of others.
- These media are used to isolate particular bacteria from a mixed culture.
- **Example:** MacConkey agar is selective for **Gram-negative bacteria**.
- **Example:** SS agar, XLD agar (Xylose Lysine Deoxycholate agar) is selective for **Salmonella** and **Shigella** species.

XLD Agar media

Salmonella



Coliforms

**Proteus
Citrobacter**



Shigella

3. Differential Media:

- Differential media contain indicators or other substances that allow the **visual distinction between different types of microorganisms** based on their biochemical properties.
- **Example:** *MacConkey agar* contains the pH indicator **neutral red**, which helps differentiate **lactose fermenting organisms** (which produce pink colonies) from **non-lactose fermenters** (which produce colorless colonies).



Escherichia coli



Enterobacter aerogenes



Proteus vulgaris



Salmonella typhimurium



Staphylococcus aureus

MacConkey's Agar

↑
**Lactose
fermenting
colonies**

PINK

↑
**Non-lactose
fermenting
colonies**

COLORLESS

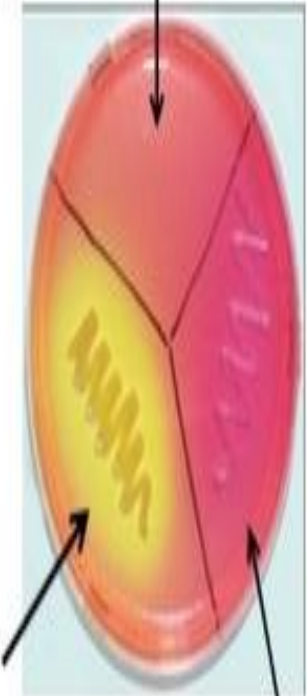
Selective and Differential Media

Mannitol Salt Agar:

Selective or differential?



*E.coli, Bacillus, Serratia,
Mycobacterium*



*Staphylococcus
aureus*

*Staphylococcus
epidermidis*

4. Transport Media:

These are specialized media used to preserve and maintain the viability of clinical specimens during transportation to the laboratory without allowing significant growth of microorganisms.

Example: Stuart's medium.



5- Anaerobic media:

These media are used to grow anaerobic organisms.



Clostridium perfringens in anaerobic blood agar



Clostridium perfringens in egg yolk agar



Bacteroides bile esculin agar

Anaerobic Culture Media



Robertson's cooked meat broth



Anaerobic phenylethyl alcohol blood agar



C. difficile in cycloserine cefoxitin fructose agar



Any Question?



Thank you